



RESEARCH PROJECT PROPOSAL

Design and Evaluation of a 3D-Printed Orthopedic Splint

Applied Engineering Club (AEC) | September–December 2026

Short-term, student-led undergraduate research project
Final presentation: physical tri-fold research board + prototype + data

Project idea in one sentence: Create a custom 3D-printed splint from arm geometry, test it against a conventional splint, and present what we learn as a small engineering research study.

Important scope: This is an Applied Engineering Club research project. It is not a clinical trial, and no prototype is to be used to treat an actual fracture.



ABSTRACT

This project asks a simple engineering question: can a custom 3D-printed splint be made lighter and more breathable than a conventional splint while still providing useful structural stiffness? Over one semester, the team will build a practical workflow that starts with arm geometry—preferably from a 3D scan—moves into CAD and lattice design, and ends with printed PETG/TPU prototypes that can be tested and compared. The project is intentionally small enough to finish by December, but technical enough to give students real experience with research: reading prior work, choosing variables, building a repeatable test, collecting data, dealing with failed prints, documenting decisions, and defending a conclusion that may or may not match the original hypothesis.

The final outcome is not expected to be a finished medical product. The goal is to produce a defensible student research result: a prototype, a small but useful dataset, and a physical tri-fold research board that explains what was tested, how it was tested, what worked, what did not, and what should be studied next.

WHY THIS PROJECT

Traditional splints are effective, but they are not designed around additive manufacturing. A 3D-printed splint gives us an approachable way to study several engineering problems at once: geometry, material behavior, stiffness-to-weight tradeoffs, manufacturing constraints, fit, ventilation, and test design. It also fits the club well because the project can involve students with different skill levels without requiring expensive equipment.

The project is also deliberately practical. We have access to school 3D printers for student use, which means we can prototype without buying a printer or paying a commercial print service. If we keep the study focused and design our test fixtures creatively, much of the work can be done for little or no money.

RESEARCH QUESTION AND WORKING HYPOTHESIS

Primary research question. How does a low-cost 3D-printed lattice splint compare with a conventional splint in mass, bending stiffness, torsional stiffness, ventilation, application time, manufacturing time, and estimated cost?

Working hypothesis. A lattice-style printed splint can reduce mass and increase open surface area while retaining enough stiffness to make the design mechanically interesting. The team should treat this as a hypothesis to test—not a conclusion to prove.

WHAT WE WILL ACTUALLY DO

- Review existing work on 3D-printed splints, orthoses, lattice structures, and low-cost mechanical testing.
- Create or obtain arm geometry. The preferred route is a 3D scan; if scanning becomes a bottleneck, an online anatomical model may be used so the project stays on schedule.
- Build a splint around the arm geometry in CAD and create at least two design variants, such as a solid/minimally open shell and a lattice shell.
- Print prototypes using school printers. PETG is the likely structural material; TPU may be explored for flexible interfaces, straps, pads, or other compliant features.
- Create simple, repeatable methods for measuring bending and torsional response. The test does not need to be sophisticated, but it does need to be documented and repeatable.
- Compare mass, stiffness, open area, print time, application time, and estimated cost.
- Analyze the results and prepare a final research board and short presentation in December.



MINIMUM VIABLE RESEARCH SCOPE

Because this is a one-semester project, the team should resist turning it into a full medical-device development effort. A successful project could be as simple as one conventional baseline plus two printed designs, tested several times with the same fixture. Good documentation and a controlled comparison are more valuable than ten flashy prototypes with inconsistent testing.

PROJECT ROLES AND RESPONSIBILITIES

Members do not need to do everything. The point is to divide the research into pieces that can move in parallel while still keeping one shared question and one shared dataset.

Area	What that team owns
Research / Literature	Find and summarize relevant papers and prior designs. Help the group decide what is worth testing and what has already been studied.
3D Scanning / Mesh	Create a usable arm mesh, clean the geometry, and document the scan process. If scanning is not practical, help source and validate an online model.
CAD / Mechanical Design	Build the splint around the arm model, create lattice variants, add closures or strap locations, and keep dimensions/parameters documented.
Additive Manufacturing	Prepare files, operate the school printers, record print settings, track failed prints, and keep material use and print time consistent.
Mechanical Testing	Design simple fixtures, define how loads will be applied, record deflection/rotation, repeat tests, and photograph the setup.
Data / Analysis	Organize measurements, calculate stiffness metrics, create graphs, compare designs, and identify limitations or sources of error.
Presentation / Documentation	Keep a running project log, collect photos and CAD screenshots, help assemble the final tri-fold research board, and coordinate the December presentation.

AVAILABLE RESOURCES AND COST EXPECTATIONS

3D printing is available at school for free student use. That is the main reason this can be a low-cost project. Students should use existing school tools, software, printers, and test equipment wherever possible before purchasing anything.

AEC may be able to provide a small amount of funding for filament, straps, hardware, or other justified project expenses. That funding should be treated as helpful, not guaranteed. If the team wants additional materials or equipment, members should be prepared to look for donations, sponsorships, campus funding, or fundraising options. Students may voluntarily spend their own money, but nobody should have to pay out of pocket in order to participate.

A good constraint for the team is to ask: “Can we answer the research question with what we already have?” In many cases, the answer should be yes. A low-cost fixture made from scrap material and a known load can still produce useful comparative data if the method is repeatable.



TIMELINE: SEPTEMBER–DECEMBER 2026

Month	Focus	Expected work	Milestone
September	Research + plan	Recruit team; review prior work; identify available printers/scanner/test equipment; find an online splint model for a quick display print; choose 2–3 designs or variables; agree on test method.	Research plan + demo print
October	CAD + prototypes	Create/obtain arm mesh; build splint in CAD; print first versions; document failures; refine fit and lattice geometry; build simple test fixtures.	First functional prototypes
November	Testing + data	Print final comparison specimens; measure mass, open area, bending response, torsional response, print time, application time, and cost; repeat tests; photograph everything.	Complete dataset
December	Analysis + presentation	Create graphs; write conclusions and limitations; print final display model; assemble tri-fold board; rehearse a short panel-style presentation and Q&A.	Board + prototype + presentation

WHAT COUNTS AS A SUCCESSFUL RESULT?

Success does not require the printed splint to “beat” a conventional splint. If the printed design is lighter but much less stiff, that is a result. If one lattice geometry performs poorly, that is a result. If the team discovers that scan-to-CAD time is the real bottleneck, that is a result too. The standard is not whether the idea wins—it is whether the experiment was fair enough that the team can explain what the data means.

FINAL PRESENTATION: PHYSICAL RESEARCH BOARD

The project will end with a physical, tri-fold/cardboard-style research presentation rather than a long slide deck. The board should be readable from a few feet away and should show the story of the project without requiring the audience to read a full paper.

LEFT PANEL	CENTER PANEL	RIGHT PANEL
Background Problem Research question Hypothesis Brief literature context	Project title + AEC Large prototype/CAD image Methodology Scan → CAD → Print → Test Photos of the build and test setup	Results and graphs Discussion Limitations Conclusion Future work

The physical prototype should be displayed with the board if practical. Team members should be ready to explain the design decisions and answer questions without reading from the board.



AI USE AND RESEARCH INTEGRITY

AEC is not treating AI as something students need to avoid. AI tools are already part of modern engineering and research, and members may use them throughout the project—for brainstorming, finding search terms, learning unfamiliar concepts, debugging code, organizing notes, checking calculations, exploring CAD approaches, or improving the clarity of their work.

The expectation is transparency and ownership. If AI materially helped with a part of the project, the team should be able to say how it was used. AI output should be checked against real sources, calculations, or experiments rather than treated as evidence by itself.

The final paper must be written by the students. AI can help the team think, outline, revise, or catch mistakes, but it should not generate the final paper for them. A rough or imperfect paper that clearly reflects the students' own understanding is better than a polished paper they cannot defend. The same applies to the research board: students should know what every graph, claim, equation, and design choice means.

This matters because the team may be asked to present the work in front of faculty, engineers, medical professionals, or other relevant reviewers. Anyone presenting should be able to explain the research question, methodology, limitations, and results in their own words and answer reasonable follow-up questions.

Good uses of AI	Not acceptable as a substitute for student work
Brainstorming Search-term generation Explaining unfamiliar concepts Code/debugging assistance Checking calculations Outlining Editing for clarity Finding possible design approaches	Inventing citations or data Presenting AI output as experimental evidence Submitting an AI-written final paper Using claims the team cannot explain Hiding material AI use from the group or audience

RESEARCH SCOPE AND SAFETY

- This is an engineering research project under the Applied Engineering Club, not a clinical study.
- Prototypes are not to be used to treat an actual fracture or replace a medically prescribed cast or splint.
- Testing should use fixtures, models, or non-injured volunteers only where appropriate and safe.
- The team should avoid claims such as “clinically equivalent,” “safe for patients,” or “medical-grade” unless those claims are actually supported by appropriate testing and oversight.
- The research should focus on engineering comparisons: geometry, stiffness, weight, ventilation, manufacturability, cost, repeatability, and limitations.

FINAL DELIVERABLES

- At least one finished 3D-printed splint prototype suitable for display.
- At least one baseline comparison and one additional printed design or configuration.
- CAD files and documented print settings.
- A small, organized experimental dataset with repeated measurements where practical.
- Graphs/tables showing the main comparisons.
- A short student-written research paper or technical report documenting the project.
- A physical tri-fold/cardboard research board for the December presentation.



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- A short oral presentation with team members prepared for questions.

The point of the project is straightforward: build something, test it carefully, keep track of what happened, and be able to explain the result.